

DIKSHA SHARMA

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PROFESSIONAL SUMMARY

Final-year BS Data Science & Applications student at IIT Madras with hands-on experience building end-to-end AI/ML systems including LLM applications, Retrieval-Augmented Generation (RAG) pipelines, NLP solutions, and full-stack AI-powered REST APIs. Proficient in Python, Scikit-learn, FastAPI, and vector databases (ChromaDB). Seeking an AI/ML internship to contribute to data-driven, production-grade AI solutions.

EDUCATION

Indian Institute of Technology Madras (IIT Madras)

BS in Data Science and Applications

Chennai, Tamil Nadu, India

July 2023 – Present (Expected 2027)

- CGPA: [Add your CGPA] | Coursework: Machine Learning, DSA, Python Programming, DBMS, Business Analytics, Linux, Java

TECHNICAL SKILLS

AI / ML: Machine Learning, Deep Learning, NLP, LLMs, RAG, Transformer Models, Feature Engineering, Model Evaluation, Model Fine-tuning

Libraries & Frameworks: Scikit-learn, PyTorch (basic), TensorFlow (basic), Hugging Face Transformers, LangChain (basic), Pandas, NumPy, Matplotlib, FastAPI, Flask, Jinja2, Bootstrap

Languages: Python, JavaScript, Java, Shell Scripting, HTML, CSS

Databases & Vector Stores: SQLite, ChromaDB, Redis

Dev Tools: GitHub, Visual Studio Code, MS Excel, Celery, REST APIs, Postman

PROJECTS

Smart Study Assistant | Python, FastAPI, OpenAI API, ChromaDB, LLMs, RAG, HTML/CSS/JS

2024

- Designed a RAG-based study assistant supporting 3 document formats (PDF, DOCX, TXT) with multi-session chat history, reducing document search time by ~60% compared to manual reading.
- Built a 5-stage NLP pipeline (extraction → chunking → embedding → indexing → retrieval) using OpenAI embeddings and ChromaDB, achieving ~85% answer relevance and sub-2s response latency.
- Developed 4 FastAPI REST endpoints for Q&A, summarization, quiz generation, and session management deployed with clean API design and error handling.

CodeInsight AI — Bug Prediction & Developer Assistant | Python, FastAPI, Scikit-learn, Radon, Pandas, NumPy, SQLite

2024

- Engineered an ML-based code analysis system that auto-extracts 5+ quality metrics (cyclomatic complexity, AST patterns, duplication proxies) to flag risky files with ML-based risk scoring.
- Automated metric extraction via Radon and AST analysis, cutting manual review time by ~40%; built scalable FastAPI APIs for real-time analysis of any Python codebase.
- Integrated a context-aware LLM chatbot and interactive dashboard delivering Code Health Scores and actionable optimization recommendations to developers.

Bank Telemarketing Campaign Prediction | Python, Scikit-learn, Pandas, Matplotlib, ML Pipelines

2024

- Built a Logistic Regression classifier on the UCI Bank Marketing dataset (45,211 samples) achieving 82% accuracy; feature engineering improved precision by 12% over the baseline classifier.
- Evaluated with Accuracy, Precision, Recall, F1-Score, and Confusion Matrix; implemented a clean sklearn Pipeline for reproducible preprocessing and end-to-end model training.

Vehicle Parking Management System | Python, Flask, VueJS, Celery, Redis, REST APIs

2024

- Built a full-stack system managing parking slot allocation, vehicle tracking, and payment computation for 50+ concurrent users with Celery-Redis async task scheduling.
- Exposed RESTful APIs for all core operations (allocation, release, billing) with real-time state updates and zero task loss under high concurrency.

CERTIFICATIONS

Diploma in Programming

IIT Madras | 2024

Diploma in Data Science

IIT Madras | 2024

Business Analytics with Excel

IIT Madras | 2024